JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR COLLEGE OF ENGINEERING (AUTONOMOUS) ANANTHAPURAMU– 515002 (A.P) INDIA
Department of Computer Science and Engineering
II B.Tech II Semester (R23)

Subject Code	Title of the Subject	L	T	P	C
23A40506	**FULL STACK DEVELOPMENT (Skill Enhancement Course)**	**0**	**1**	**2**	**2**

Course Objectives: The main objectives of the course are to

- Understand the Fundamentals of Web Programming.
- Apply Coding Standards and Conventions.
- Apply CSS Styling Techniques, and properties to style web pages effectively, including color manipulation, font styling, and element positioning.
- Ability to Organize Web Page Content and build static web pages effectively.
- Build a web page by applying appropriate CSS styles to HTML elements
- Implement object oriented programming with JavaScript to develop dynamic web pages and validate forms

Course Outcomes: Students can
CO1: Design Websites. (L6)
CO2: Apply Styling to web pages. (L3)
CO3: Develop interactive Web pages. (L6)
CO4: Design Forms for applications. (L6)
CO5: Apply JavaScript control structures ensuring effective and logical code execution. (L3)
CO6: Implementation of real time application on Advanced Java Script Concepts

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO1 0**	**PO1 1**	**PO1 2**	**PSO 1**	**PSO 2**	**PSO 3**
CO 1	1	2	1											2	
CO 2														2	
CO 3		1			2									2	
CO 4			3	1	2	2								2	
CO 5	1										1		1	2	
CO 6												1		2	

Chapter 1:
Introduction to Web Programming: Introduction, Creating a Website, Web Page Example, HTML Tags, Structural Elements, title Element, meta Element, HTML Attributes, body Elements hr, p, br, div, Cascading Style Sheets Preview, History of HTML, HTML Governing Bodies, Differences Between Old HTML and HTML5.
Coding Standards, Block Elements, Text Elements, and Character References: Introduction, HTML Coding Conventions, Comments, HTML Elements Should Describe Web Page Content Accurately, Whitespace Collapsing, pre Element, 0 Phrasing Elements, Editing Elements, q and cite Elements, dfn, abbr, and time Elements, Code-Related Elements, br and wbr Elements, sub, sup, s, mark, and small Elements, strong, em, b, u, and i Elements, span Element, Character References, Web Page with Character References and Phrasing Elements, Forms and Form Elements.
Programs:

Task - 1. Lists, Links and Images

a. Write a HTML program, to explain the working of lists.
   Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.
b. Write a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.
c. Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
d. Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

Task - 2. HTML Tables, Forms and Frames

a. Write a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)
b. Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
c. Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).
d. Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame paragraph, third frame hyperlink. And also make sure of using "no frame" attribute such that frames to be fixed).

Chapter 2:

Cascading Style Sheets (CSS): Introduction, CSS Overview, CSS Rules, Example with Type Selectors and the Universal Selector, CS5 Syntax and Style, Class Selectors, Selectors, span and div Elements, Cascading, style Attribute, style Container, External CSS Files, CSS Properties, Color Properties, RGB Values for Color, Opacity Values for Color, HSL and HSLA Values for Color, Font Properties, line-height Property, Text Properties, Border Properties, Element Box, padding Property, margin Property.

Organizing a Page's Content with Lists, Figures, and Various Organizational Elements: Introduction, Unordered Lists, Descendant Selectors, Ordered Lists, Organizational Elements, section, article, and aside Elements, nav and a Elements, header and footer Elements, Child Selectors, CSS Inheritance.

Programs:

Task - 3. HTML 5 and Cascading Style Sheets, Types of CSS

a. Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, , tags.
b. Write a HTML program, to embed audio and video into HTML web page.
c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value).

Task - 4. Selector forms

a. Write a program to apply different types of selector forms
   i. Simple selector (element, id, class, group, universal)
   ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
   iii. Pseudo-class selector
   iv. Pseudo-element selector
   v. Attribute selector

Chapter 3:

Tables and CSS Layout: Introduction, Table Elements, formatting a Data Table Borders, Alignment, and Padding, CSS Structural Pseudo-Class Selectors, thread and tbody Elements, Cell Spanning, Web Accessibility, CSS display Property with Table Values, Absolute Positioning with CSS FotoProperties, Relative Positioning.

Links and Images: Introduction, an element**,** Relative URLs, index.html File, Web Design, Navigation Within a Web Page, CSS for Links, an Element Additional Details, Bitmap Image Formats: GIF, JPEG, PNG, img Element, Vector Graphics.

Image Manipulations, Audio, and Video: Introduction, Positioning Images, Shortcut Icon, iframe Element, CSS Image Sprites, Audio, Background Images, Web Fonts, Video.

Programs:

Task - 5. CSS with Color, Background, Font, Text and CSS Box Model

a. Write a program to demonstrate the various ways you can reference a color in CSS.
b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
c. Write a program using the following terms related to CSS font and text:
   i. font-size ii. font-weight iii. font-style
   iv. text-decoration v. text-transformation vi. text-alignment
d. Write a program, to explain the importance of CSS Box model using
   i. Content ii. Border iii. Margin iv. padding

Task - 6. Applying JavaScript - internal and external, I/O, Type Conversion

a. Write a program to embed internal and external JavaScript in a web page.
b. Write a program to explain the different ways for displaying output.
c. Write a program to explain the different ways for taking input.
d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

Chapter 4:

Introduction to JavaScript: Functions, DOM,Form and Event Handlers:

Introduction, History of JavaScript, Hello World Web Page, Buttons, Functions, Variables, Identifiers, Assignment Statements and Objects, Document Object Model, Forms and How They're Processed Client-Side Versus Server-Side, form Element, Controls, Text Control, Email Address Generator Web Page, Accessing a Form's Control Values, reset and focus Methods, Comments and Coding Conventions, Event-Handier Attributes, onchange, onmouseover, onmouseout, Using noscript to Accommodate Disabled JavaScript.

Loops, Additional Controls, Manipulating CSS with JavaScript: Introduction, while Loop, External JavaScript Files, Compound Interest Web Page, do Loop, Radio Buttons, Checkboxes, Job Skills Web Page, for Loop, fieldset and legend Elements, Manipulating CSS with JavaScript, Textarea Controls, Dormitory Blog Web Page, Pull-Down Menus, List Boxes.

Programs:

Task - 7. JavaScript Pre-defined and User-defined Objects

a. Write a program using document object properties and methods.
b. Write a program using window object properties and methods.
c. Write a program using array object properties and methods.
d. Write a program using math object properties and methods.
e. Write a program using string object properties and methods.
f. Write a program using regex object properties and methods.
g. Write a program using date object properties and methods.
h. Write a program to explain user-defined object by using properties, methods, accessors, constructors and display.

Task - 8. JavaScript Conditional Statements and Loops

a. Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words "LARGER NUMBER" in an information message dialog. If the numbers are equal, output HTML text as "EQUAL NUMBERS".
b. Write a program to display week days using switch case.
c. Write a program to print 1 to 10 numbers using for, while and do-while loops.
d. Write a program to print data in object using for-in, for-each and for-of loops
e. Develop a program to determine whether a given number is an 'ARMSTRONG NUMBER' or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e.,13 + 53+ 33 = 153]
f. Write a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1- 10's, 1-2's & 1-1's)

Chapter 5:

Additional JavaScript Basics: window Object, if Statement, Strings, Numbers, and Input Validation: Introduction, window Object, alert and confirm Methods, if Statement (E by itself, Game Night Web Page, prompt Method, Game Night Wath Page Revisited, if Statement else and else if Clauses, Strings, Word Ordering web page, Arithmetic Operators, Parsing Numbers, parseInt, parseFloat,

Object-Oriented Programming and Arrays: Introduction, Object-Oriented Programming Overview, Classes, Constructors, Properties, new Operator, Methods, Point Tracker Web Page, static Methods, Event Handlers, Primitive Values Versus Objects, Using addEventListener to Add Event Listeners, Using Prototypes to Emulate a Class, Inheritance Between Classes, Pet Registry Web Page, switch Statement, Arrays, Arrays of Objects.

Programs:

Task - 9. JavaScript Functions and Events

a. Design a appropriate function should be called to display
    i. Factorial of that number
    ii. Fibonacci series up to that number
    iii. Prime numbers up to that number
    iv. Is it palindrome or not
b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
    i. Factorial of that number
    ii. Fibonacci series up to that number
    iii. Prime numbers up to that number
    iv. Is it palindrome or not
c. Write a program to validate the following fields in a registration page
    i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
    ii. Mobile (only numbers and length 10 digits)
    iii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

Task - 10. Java Script Object oriented programming

a. Write a function to that takes two numbers and returns smallest of two, or square of two if they are equal and demonstrate js scope variables

b. Write JS function to show Local Variables Have Priority Over Global Variables.

c. Write a java script program to convert day of week number to day of week name using closures

d. Write a java script program to guess the password and validate. If the password is not correct display the security question. Use closures to implement the same.

e. Write a JavaScript program to delete the rollno property from the following object. Also print the object before or after deleting the property.

```
var student = {
    name : “Ram Mohan",
    Sem : "VI",
    rollno : 62 };
```

f. Create an object to hold information on your favorite recipe. It should have properties for title (a string), servings (a number), and ingredients (an array of strings). On separate lines (one console.log statement for each), log the recipe information so it looks like:
   - Mole
   - Serves: 2
   - Ingredients:
   - cinnamon
   - cumin
   - cocoa

g. Write a js program using prototype to calculate square, cube and square root of a number.

h. Write a Calculator class using prototype, which has add, subtract and getAnswer functions. The class should support chaining, that means we should be able to do new Calculator(2).add(2).add(2).subtract(3).getAnswer() to get 3 as the answer.

i. Write an Animal class with methods: makeSound and run. It should take parameters: name, sound, speed (in meter per second). The run method takes a parameter called time and returns a Promise which resolves with the distance the animal ran after “time” seconds and the name of the animal. (hint: use setTimeout inside a Promise). Construct 5 different animals with different speeds, store them in an array, call their run method with 5 seconds in a loop, wait for all the animals to finish running and then log which animal went the farthest

Text Books:

1. John Dean, “Web Programming-With HTML5, CSS and JavaScript”,Jones and Bartlett learning.

Reference Books:

1. Douglas Crockford , "JavaScript: The Good Parts”, Oreilly Publication.
2. Jennifer Robbins, "Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics",5th Edition, Oreilly Publication.
3. Terry Felke-Morris, "Web Development and Design Foundations with HTML5", 10th edition, Pearson.
4. Robin Nixon, "Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5", 4th Edition, Oreilly.